

Enhanced Multi-Scale Graph Networks for Glaucoma Detection in Optical Coherence Tomography

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Abstract— Precise and automated tissue segmentation plays a vital role in diagnosing glaucoma using ocular optical coherence tomography (OCT) images. The intricate anatomical structure of the peripapillary region, combined with the existence of the optic disc, contributes to the difficulty of the task. In order to fix this challenge, we have created a cutting-edge two-stage architecture that uses a convolutional graph network (GCN) to identify both the ocular disc and nine ocular layers concurrently. This method incorporates a multifaceted spatial processing module into a U-shaped neural network, leveraging anatomical previous knowledge to strengthen spatial thinking skills. The module is placed between the encoder and decoder portions of the network. Our segmentation network achieved a Dice score of 0.850 ± 0.0012 and a pixel accuracy of 0.895 ± 0.0019 , outperforming existing state-of-the-art techniques.

Keywords - Glaucoma, Tissue segmentation, Graph convolutional network, U-shaped neural network, Multi-scale global reasoning, Peripapillary region, Biomedical image processing, Dice score, Pixel accuracy

I. INTRODUCTION

Glaucoma affects over 64.3 million people globally and continues to be the primary cause of permanent vision loss [1]. In 2015, approximately 13.12 million individuals in China had glaucoma, with estimates suggesting this number could increase to 25.16 million by 2050 [2]. This projected rise will place significant pressure on the public health system. Early detection and effective management are critical to preventing vision loss from glaucoma. Accurate diagnosis largely depends on identifying subtle changes in retinal layers, such as thinning of the cell layer and ocular nerve fiber layer (RNFL) [3]. OCT, a non-invasive 3D imaging technology, is extensively utilized in ophthalmology clinics to assess the retina. OCT's micrometer-level axial resolution allows for the direct observation and measurement of the retina's layered structure. Peripapillary RNFL thickness, measured by OCT, is commonly used to diagnose early-stage glaucoma [4]. Hence, accurate delineation of tissue regions in retinal OCT images is crucial for efficient early detection of glaucoma. Manual segmentation requires a significant amount of time and effort, highlighting the need for an accurate automated system to benefit researchers and clinicians alike. Over the past few decades, numerous automated methods for segmenting retinal OCT images have been developed [5]. After image preprocessing, Alonso-Caneiro et al.

and Tian et al. employed graph based edge analyzers to accurately detect the choroidal margin [6]. In addition, Chiu et al. presented a segmentation technique based on kernel prediction for ocular OCT images that are impacted by macular edema caused by diabetes [7]. For instance, OCTExplorer employs a traditional graph-theory approach to extract retinal boundaries [8]. Lang et al. created an algorithm using random forests to divide macular cube images into eight retinal layers [9]. Mayer et al. introduced an energy reduction module for segmenting the surface of the ocular nerve fiber layer in OCT images [10]. This study introduces our new investigation that aims to utilize the historical data included within ocular OCT images. We affirm that the structural arrangement of all peripapillary OCT images, acquired adhering to a stringent clinical protocol, demonstrates a uniform pattern: The optic nerve head is a conspicuous anatomical feature, typically resides centrally within the image, flanked by thinner retinal layers on either side, as depicted in Fig. 1. In order to effectively utilize these existing anatomical assumptions, we have created a unique two-stage network that incorporates a variable-scale graph convolutional network. This framework is specifically developed to perform simultaneous sectioning of ocular layers and the disc of optics in ocular OCT images. The methodology we employ is influenced by the technique suggested by Jamal et al., which employs graph-based representation to capture domain knowledge and structural relationships among tissues.

Experiments were conducted using a curated dataset of peripapillary OCT scans, comprising 122 B-scans from 71 patients, alongside an additional publicly available dataset, demonstrating the effectiveness of our proposed framework. Comparative evaluations with baseline methods and state-of-the-art techniques revealed superior performance across both datasets. Moving forward, our goal is to integrate this segmentation framework into diagnostic procedures aimed at early-stage glaucoma detection.

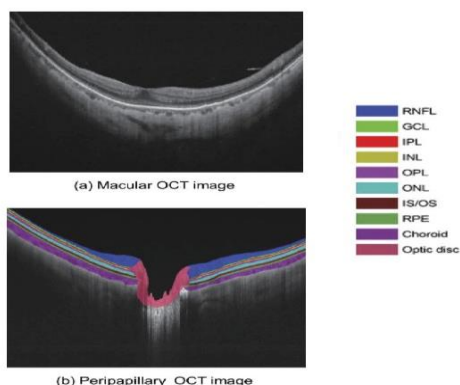


Fig 1. (a) Macular zone OCT and (b) Ocular OCT

Numerous techniques have been devised to enhance the precision and effectiveness of ocular OCT image sectioning and noise mitigation, which are crucial for early glaucoma detection. One prominent example is OCT Explorer, which uses a traditional graph-theory approach for extracting retinal boundaries [11]. Alonso-Caneiro et al. and Tian et al. employed graph-based border analyzers after image preprocessing to delineate the choroidal margin [12]. Mayer et al. initiated a reduction method to segment the surface of the layer of retinal nerve fibers in cyclic OCT images [13]. Lang et al. developed a categorization model to divide myopic cubed images into eight ocular layers. This approach significantly improved the accuracy of segmentation, as reported in their study [15]. Chiu et al. introduced a technique for segmentation based on base regression analysis specifically designed for ocular OCT images affected by macular edema caused by diabetes, proving its effectiveness in addressing complex scenarios [15].

In addition to segmentation, several noise reduction techniques have been explored. The Block-Based Autoencoder Network (BBAuto-Net) and Juneja et al.'s Bayes Shrinkage based Fused Wavelet Transform (BSbFWT) are notable methods for reducing noise in MRI images [6]. These techniques involve fusing and reconstructing wavelet transforms such as Haar, Symlet, and Daubechies, followed by applying fusion techniques to enhance image quality. The BM3D (Block Matching and 3D Filtering) algorithm by Dabov et al. is another effective denoising method that operates on the principle of collaborative filtering, grouping similar image patches for joint filtering to reduce noise [7]. Rawat et al.'s CV-CNN (Complex-Valued Convolutional Neural Network) leverages complex-valued numbers in convolutional layers to capture more intricate patterns compared to traditional CNNs [8]. Dazi Li et al.'s Structural Convolutional Neural Network (SCNN) utilizes multiple convolutional layers and a unique feature fusion process to effectively mitigate speckle noise in medical images [9]. Wang et al. introduced the HVSr (Hybrid Variation Sparse Representation) algorithm, which dissects medical images into structure, detail, and noise layers, and applies a sparse representation-based approach for enhanced denoising [10]. Lastly, Chervyakov et al. studied the impact of quantization noise on the performance of Discrete Wavelet Transforms (DWT) filters and proposed Distributed Arithmetic (DA), an efficient method for implementing mathematical operations using elementary shift and add operations, suitable for hardware implementation [11].

II. PROPOSED METHOD

The current study involves manually labeling 10 markers on the OCT dataset to assist in the training process. The annotation process is illustrated in Figure 1(b).

A. Summary of the segmentation model

The network for detecting the macular disc and the system for segmenting ocular layers are precisely constructed to utilize the structural knowledge of the ocular region and overcome the related difficulties of segmenting retinal layers of varying thicknesses, as depicted in Fig 3. The right side of Fig 3 illustrates a horizontal configuration of the disc and non-disc regions on a global level, following a pattern where non-disc areas alternate with disc areas. Conversely, the left panel of Fig 3 focuses on the "non-disc" area, revealing vertically stacked retinal layers of different thicknesses. The optic disc detection network provides extensive spatial reasoning in a horizontal direction, the ocular layer recognition network catches the various vertical structures at different levels.

B. Graph inference module

This method is a feature extraction framework combining projection, graph convolution, inverse projection, and feature fusion to enhance the representation of original features (X). In the Projection phase (I), the original features are reshaped into different matrices (X_r and X_a). The reshaped features (X_r) and projection matrix (X_a) undergo Graph Convolution (II), where graph convolutional network (GCN) blocks extract node and global node features (H and Z). During Inverse Projection (III), the global node features (Z) are inversely projected and reshaped to obtain new features (X_d). In the Fusion phase (IV), the global features (M) are combined with the inversely projected features (X_d) to form the final enhanced features (Y) through convolution and addition operations. This method aims to leverage graph-based relationships to improve the robustness and expressiveness of the original features as given in Fig 2.

C. Multi-scale global inference unit

This method is a multi-scale feature extraction approach designed to enhance the representation of input features through a combination of convolution, pooling, graph reasoning, and interpolation. The input features first pass through a 3×3 convolution layer, after which they are subjected to different scales of max-pooling (2×2 , 3×3 , and 5×5) along with a branch

that bypasses pooling. Each branch processes the features through a graph reasoning block, enabling the network to discern relationships between features across different scales. After graph reasoning, the features from each branch are interpolated to match the original input dimensions. Finally, all branches'

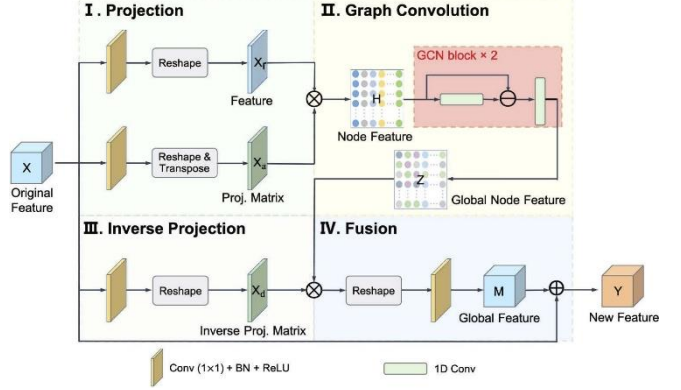


Fig 2. The framework of a graph reasoning block

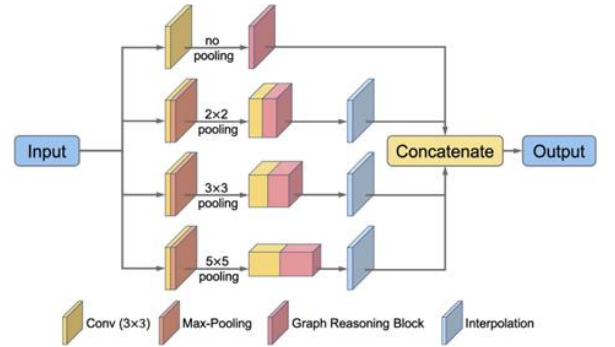


Fig 3. The architecture of the multi-scale global inference unit

III. EXPERIMENTAL SETUP

A. DATASETS AND IMPLEMENTATION DETAILS

In this research, we carried out experiments on a dataset of ocular OCT images, which were recognized and approved by the Mahatma Eye Hospital Private Limited, Tiruchirappalli. The dataset consists of 71 subjects, with each subject having 14 radial OCT B-scans acquired. These images have dimensions of 224 x 224 pixels, equivalent to a field of point of 22.48 mm x 7.94 mm. Two graders have been allocated to demarcate the lens of the eye and the nine compartments of the retina, supervised by a glaucoma specialist, who manually annotated each subject's data. An inter-grader consensus approach was employed to establish the ground truth. The collection of data was dynamically partitioned into training, validation, and testing on segments with a mean of 70:10:20 for each subject. Various data augmentation approaches, such as horizontal flipping over, additive Gaussian fluctuations, and brightness modifications, were applied to expand the dataset.

Furthermore, we evaluated our proposed methodology by utilizing the Duke SD-OCT data by Chiu et al., which comprises 110 OCT B-scans from ten individuals with diseases from diabetes, with an image dimensions of 496 × 768 pixels.

B. EVALUATION STUDY

In order to evaluate the particular benefits of each distinct element in our planned design, we performed several elimination tests utilizing the gathered dataset. The assessment entailed a comparison of the suggested model's performance with that of four different versions: (1) a single-stage baseline model, (2) a two-stage baseline model, (3) a model incorporating a single-scale global inference unit within a structure of a pair segmentation, and (4) an algorithm without graph assessment elements in the numerous scales global reasoning unit.

C. ASSESSMENT METRICS

The Dice Score (DS) and Pixel Accuracy (PA) were utilized to quantitatively assess segmentation performance by comparing predictions with reference segmentations.

$$DS = \frac{2|A \cap B|}{|A| + |B|} \quad (1)$$

$$PA = \frac{|A \cap B|}{|B|} \quad (2)$$

IV. RESULTS AND DISCUSSION

The suggested approach shows enhanced performance across most macular nerve tissue division when compared to selected advanced methods, with the exception of the ocular disc (both Dice score and pixel accuracy) and RPE (pixel accuracy). On average, the proposed network achieves Dice scores that are 1.9%, 2.6%, and 2.3% higher than ReLayNet, U-Net, and DRUNET, respectively. Table 2 details the empirical results from the collected dataset, with similar trends seen in pixel accuracy metrics. Figure 4 depicts the segmentation results produced by different methods on a typical peripapillary OCT image. The segmented image generated by our proposed technique (Figure 4(f)) shows significantly better visual quality compared to others, which exhibit noticeable artifacts. These unwanted details can be classified into two main types: 1) Layer discontinuities, where retinal layers abruptly end horizontally, visible in ReLayNet (Figure 4(d)) and DRUNET (Figure 4(e)), marked by white stars. 2) Scattered labels, referring to isolated regions mistakenly identified as belonging to different layers, are indicated by yellow arrows in Figures 4(c)-(e). Such errors are present in all methods except our proposed approach. The enhanced performance of our method can be attributed to the incorporation of additional anatomical information within the framework. Traditional CNN-based algorithms often rely heavily on textural details for pixel classification. In contrast, our approach imposes spatial constraints that ensure segmentation results conform to learned spatial structures, requiring horizontally continuous and vertically stratified arrangement of retinal layers.

TABLE 1. CLINICAL PROFILE OF SUBJECTS IN THE DATASET

Parameters	Value
Scan	71
Male, n (%)	28 (46.8%)
Age in yrs, mean ± standard deviation	68.20 ± 5.71
Ocular pressure, mean ± standard deviation	13.45 ± 2.09
Length of axis, mean ± standard deviation	28.43 ± 1.69
Increased myopia, x (%)	32 (54.70%)
Atrophy, x (%)	37 (59.12%)
Cataract, x (%)	31 (48.42%)

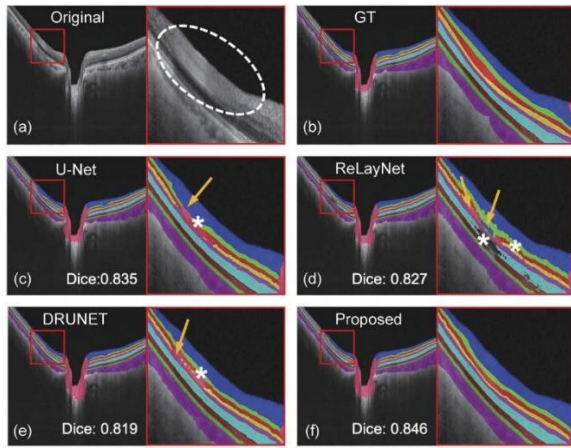


Fig 4. Segmentation results. Yellow arrows highlight scattered labels, while white stars indicate layer discontinuities. Magnified views are included for enhanced clarity

TABLE II. SEGMENTATION RESULTS OBTAINED FROM VARIOUS APPROACHES ON THE COLLECTED DATASET

	U-Net	DRUNET	ReLayNet	Proposed
IPL	71.0 $\pm$ 0.9	72.5 $\pm$ 1.1	72.7 $\pm$ 1.2	75.7 $\pm$ 0.6
INL	76.1 $\pm$ 0.7	76.5 $\pm$ 1.1	77.2 $\pm$ 1.3	78.9 $\pm$ 0.5
OPL	77.2 $\pm$ 0.8	80.0 $\pm$ 0.9	81.2 $\pm$ 0.7	82.3 $\pm$ 0.4
ONL	91.6 $\pm$ 0.1	90.8 $\pm$ 0.3	91.0 $\pm$ 0.2	91.1 $\pm$ 0.1
IS/OS	86.7 $\pm$ 0.3	85.9 $\pm$ 0.3	86.5 $\pm$ 0.2	86.2 $\pm$ 0.2
RPE	81.2 $\pm$ 0.4	82.4 $\pm$ 0.5	85.3 $\pm$ 0.7	83.9 $\pm$ 0.3
Choroid	87.4 $\pm$ 0.3	86.3 $\pm$ 0.7	89.3 $\pm$ 0.4	88.4 $\pm$ 0.4
Disc	8423 $\pm$ 0.8	77.2 $\pm$ 0.8	7732 $\pm$ 1.4	82.1 $\pm$ 0.7
RNFL	83.7 $\pm$ 0.6	82.4 $\pm$ 0.3	81.7 $\pm$ 0.3	83.1 $\pm$ 0.3
GCL	68.5 $\pm$ 1.7	69.6 $\pm$ 0.7	69.3 $\pm$ 0.7	70.5 $\pm$ 0.3

V. CONCLUSION AND FUTURESCOPE

By embedding a multi-scale global reasoning module into a U-shaped neural network design, we are able to efficiently exploit the anatomical information that is already available, as well as handle the obstacles that are accommodating the various sizes and the morphology of the disc of vision and ocular layers. With the help of a two-stage framework, this module makes it possible to do thorough long-range spatial reasoning, which is necessary for the precise segmentation of optic discs and retinal layers in peripapillary OCT. Our study on a dataset that has been curated as well as a dataset that is accessible to the public indicates a considerable improvement in

segmentation performance when compared to approaches that are considered to be state-of-the-art. To be more specific, the method that we have developed is capable of achieving an average Dice score of 84.0% and pixel accuracy of 86.0%, which is 1.9% and 2.9% higher than ReLayNet, correspondingly accordingly. The enhanced visual clarity of segmented pictures demonstrates how well anatomical information can be integrated via the use of the multi-scale global reasoning module and demonstrates its usefulness. Utilizing this segmentation network's potential for additional biomedical image segmentation tasks that need rich anatomical priors is our long-term objective. We aim to incorporate this segmentation network into clinical workflows for the early-stage diagnosis of glaucoma diagnosis. In addition, we want to contribute to the field by making our segmentation dataset and the scripts connected with it available to the general public. This might further research in peripapillary retinal optical coherence tomography picture reconstruction. Moving forward, this research paper holds several promising avenues for future exploration. Firstly, there is a significant opportunity to integrate the proposed segmentation network into clinical practice for automated or real-time diagnosis of early-stage glaucoma, refining its performance on larger, more diverse datasets to enhance its utility in routine clinical workflows.

Secondly, ongoing development of the multi-scale global reasoning module can expand its capability to incorporate more detailed anatomical priors, thereby improving segmentation accuracy across varied anatomical variations and pathologies. Additionally, exploring the adaptability of the framework to other biomedical imaging tasks beyond OCT could broaden its impact in medical imaging diagnostics. Integrating the segmentation network into AI-assisted decision support systems represents another avenue, leveraging combined clinical data for comprehensive ophthalmic diagnostics. Continuing efforts to expand and refine the segmentation dataset will ensure its ongoing usefulness for benchmarking and advancing research in peripapillary retinal OCT image analysis. Further algorithmic refinements, validation studies, and clinical trials will be crucial in validating and optimizing the network's performance for diverse patient populations and clinical settings, potentially leading to broader adoption in clinical practice.

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